

Tutorial wk 5

Worked example- Deflection calculation for simply supported beam

Beam shown below spans 20 m and supports a live load of 14.8 kN/m in addition to self weight.

(a) Check the adequacy of the beam at transfer using the provisions of Cl. 8.1.6.2

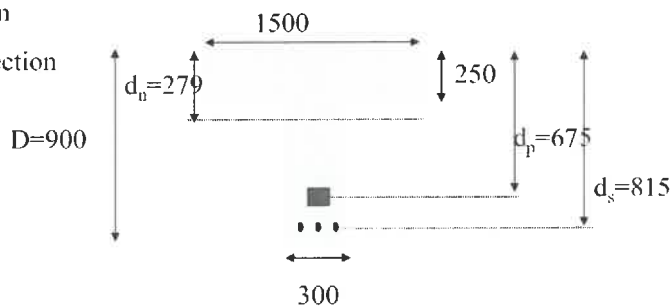
(b) Calculate the deflection of the beam.

-short term deflection at transfer

-short term deflection under service loads

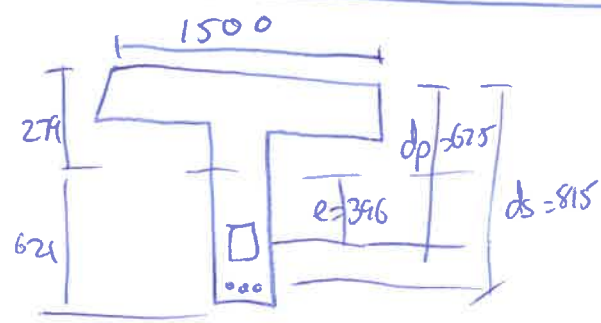
-creep deflection

-long term deflection



$$\begin{aligned} f'_{cp} &= 35 \text{ MPa} \\ f'_{cs} &= 45 \text{ MPa} \\ P_c &= 1590 \text{ kN} \\ P_i &= 1749 \text{ kN} \\ A_p &= 1580 \text{ mm}^2 \\ A_s &= 1350 \text{ mm}^2 \\ A_u &= 570,000 \text{ mm}^2 \\ I_g &= 3.48 \times 10^{10} \text{ mm}^4 \\ I_{cr} &= 8.58 \times 10^9 \text{ mm}^4 \\ E_c &= 33800 \text{ MPa} \\ E_p &= 200000 \text{ MPa} \\ Z_{hut} &= I_g / y_{hut} \\ &= 3.48 \times 10^{10} / 621 \\ &= 56.04 \times 10^6 \text{ mm}^3 \end{aligned}$$

Week 5- Tutorial Problem



span = 20m

concrete : 25 kN/m^3
specific weight

$$f'_c = 45 \text{ MPa}$$

$$f'_{cp} = 32 \text{ MPa}$$

$$P_e = 1590 \text{ kN}$$

$$P_i = 1749 \text{ kN}$$

$$E_{cp} = 30100 \text{ MPa}$$

$$E_c = 33800 \text{ MPa}$$

$$I_g = 3.48 \times 10^{10}$$

$$A_g = 570,000$$

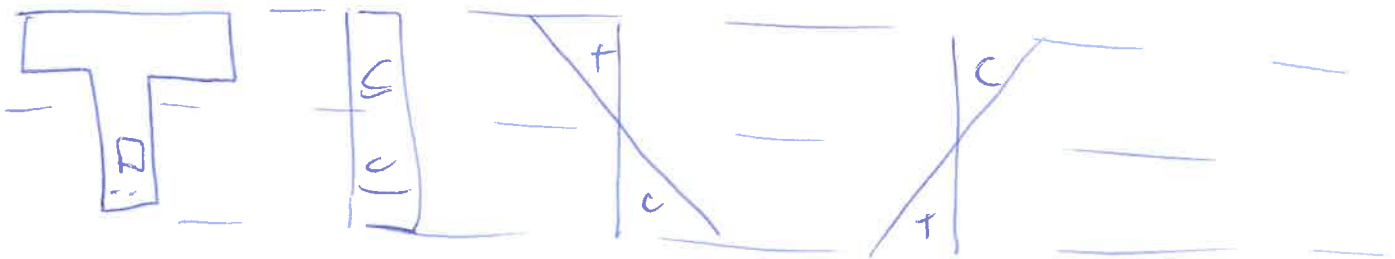
$$A_p = 1580 \text{ mm}^2$$

$$A_s = 1350 \text{ mm}^2$$

a) @ transfer beam self-weight load

$$W_g = 25 \text{ kN/m}^3 \times 0.57 \text{ m}^2 = 14.3 \text{ kN/m}$$

$$M_g = \frac{w_l^2}{8} = \frac{14.3 \times 20^2}{8} = 713 \text{ kNm}$$



$$\sigma_b = \frac{P_i}{A_g} + \frac{P_i \cdot e \cdot y_b}{I_g} - \frac{M_g \cdot y_b}{I_g}$$

$$\sigma_b = \frac{1749 \times 10^3}{570,000} + \frac{1749 \times 10^3 \times 396 \times 621}{3.48 \times 10^{10}} - \frac{713 \times 10^6 \times 621}{3.48 \times 10^{10}}$$

$$= 2.6 + 12.4 - 12.7$$

$$= 2.3 \text{ MPa} < 0.6 f_{cp} (0.6 \times 32 = 19.2 \text{ MPa})$$

= ok.

b)(i) short-term deflection @ transfer.

- Equivalent upward load due to prestress

$$w_p = \frac{8 \cdot P_i \cdot h}{L^2} = \frac{8 \times 1590 \times 0.396}{20^2}$$
$$= 13.9 \text{ kN/m } \uparrow$$

$$w_g = 14.3 \text{ kN/m } \downarrow$$

- Resultant net load

$$w_g - w_p = 14.3 - 13.9 \uparrow = 0.4 \text{ kN/m } \downarrow$$

$$\Delta_{ST} \text{ @ transfer} = \frac{5}{384} \frac{w_{net} \cdot L^4}{E_c I_g}$$
$$= \frac{5}{384} \times \left(\frac{0.4 \times 20,000^4}{30,100 \times 3.48 \times 10^{10}} \right)$$
$$= 0.8 \text{ mm } \downarrow$$

(ii) short-term deflection under service loads.

$$P_e = 1590 \text{ kN}$$

$$w_{service} = w_g + \psi_{ST} w_q$$

$$\psi_{ST} = 0.7$$

$$= 14.3 + 0.7 (14.8)$$

$$= 24.7 \approx 25 \text{ kN/m}$$

$$M_{service} = \frac{w l^2}{8} = \frac{25 \times 20^2}{8} = \underline{1250 \text{ kNm}}$$

* need to check if beam is cracked?

$$\text{Calculate } M_{cr} = Z \left(f_{ctf} - \sigma_{cs} + \frac{P_e}{A_g} \right) + P_e \cdot e$$

→ Cl 8.5.3.1
of AS3600

$$f'_{cf} = 0.6 f'_c$$

$$= 0.6 \sqrt{45} = 4 \text{ MPa}$$

$$\epsilon^A_{cs} = \epsilon_{cs} = 0.00049 \quad (f'_c = 45, t = 30 \text{ yd}, t_h = 238 \text{ mm})$$

$$p_w = \left(\frac{A_{st} + A_p}{b w d} \right) = \left(\frac{1350 + 1580}{300 \times 45} \right) = 0.013$$

$$\sigma_{cs} = \text{maximum shrinkage induced tensile stress}$$

$$= E_s \epsilon^A_{cs} \left(\frac{2.5 p_w - 0.8 p_{cw}}{1 + 50 p_w} \right) \quad p_{cw} = 0$$

$$\sigma_{cs} = 1.73 \text{ MPa}$$

$$Z = \frac{I_g}{y_b} = \frac{3.48 \times 10^{10}}{621} = 5.6 \times 10^7 \text{ mm}^3$$

$$\Rightarrow M_{cr} = 5.6 \times 10^7 \times \left(4 - 1.73 + \frac{1590 \times 10^3}{570 \times 10^3} \right) +$$

$$(1590 \times 10^3 + 396)$$

$$M_{cr} = 1113 \times 10^6 \text{ Nmm}$$

$$= 1113 \text{ kNm} < M_{\text{service}} (1250 \text{ kNm})$$

$\therefore$ the beam section @ midspan
is cracked.

Hence $I_{ef} \neq I_g$.

Effective second moment of area is

$$I_{ef} = I_{cr} + \left(I_g - I_{cr} \left(\frac{M_{cr}}{M_s} \right)^3 \right) \leq I_{ef, max}$$

$\Rightarrow$ now need to calculate second moment of area of a cracked section I_{cr} .

* Calculation of neutral axis depth and second moment of area of partially prestressed concrete section

$$M_{service} = 1250 \text{ kNm}$$

$$P_e = 1590 \text{ kN}$$

$$d_p = 675 \text{ mm}$$

$$A_p = 1880 \text{ mm}^2$$

$$A_{st} = 1350 \text{ mm}^2$$

$$d_{st} = 845 \text{ mm}$$

$$E_c = 33800 \text{ MPa}$$

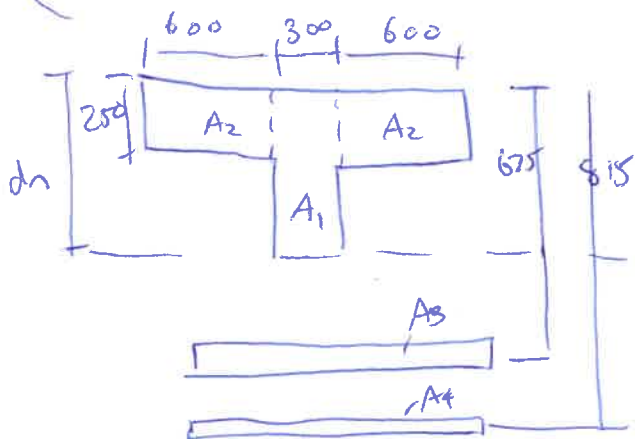
$$E_s = E_p = 200 \times 10^3 \text{ MPa}$$

$$A_g = 570,000 \text{ mm}^2 \quad I_g = 3.48 \times 10^{10} \text{ mm}^4$$

$$y_{top} = 279 \text{ mm}$$

— first determine d_n by equating moments above and below neutral axis. Consider transformed section and assume d_n is in the web.

(use modular ratio + assume tendons are prestress for purpose of calculating d_n & I_{cr})



$$n = \frac{E_s}{E_c} = 5.9$$

$$A_1 = 300 d_n$$

$$A_2 = 1200 \times 250 = 300,000 \text{ mm}^2$$

$$A_3 = n A_p = 5.9 \times 1880 = 9322 \text{ mm}^2$$

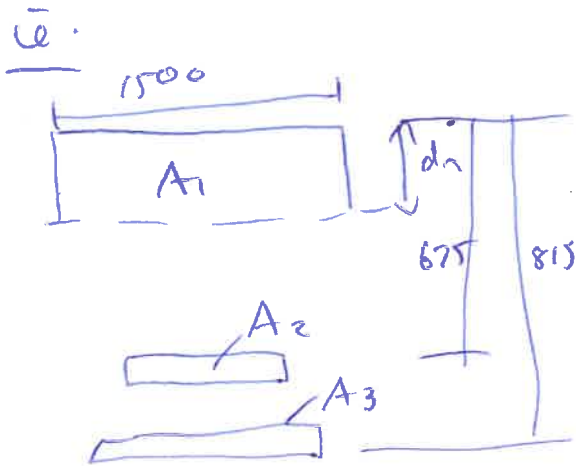
$$A_4 = n A_{st} = 5.9 \times 1350 = 7965 \text{ mm}^2$$

$\sum M$ about ~~area~~ neutral axis ($\sum A y = 0$)

$$\left(300 d_n \times \frac{d_n}{2} \right) + \left[300,000 \times (d_n - 125) \right] - \left[9322 \times (675 - d_n) \right] - \left[7965 \times (815 - d_n) \right] = 0$$

$$150 d_n^2 + 317287 d_n - 50,283,825 = 0$$

$d_n = 149 \text{ mm}$ $\therefore d_n$ in flange not web, need to re-calculate.



$$A_1 = 1500 \cdot d_n$$

$$A_2 = n A_p = 9322 \text{ mm}^2$$

$$A_3 = n A_s = 7965 \text{ mm}^2$$

$\sum M$ about d_n .

$$\left(1500 \times d_n \times \frac{d_n}{2} \right) - \left[9322 \times (675 - d_n) \right] - \left[7965 \times (815 - d_n) \right] = 0$$

$$750 d_n^2 + 17287 d_n - 12,783,825 = 0$$

$d_n = 119.8 \approx \underline{120 \text{ mm}}$ in flange.

* Determine second moment of area of cracked section @ neutral axis, d_n $(\Sigma I + A_y^2)$

$$I_{cr} = \frac{1}{12} \times 1500 d_n^3 + 1500 \times d_n \left(\frac{d_n}{2} \right)^2$$

$$+ 9322 (d_p - d_n)^2$$

$$+ 7965 (d_s - d_n)^2$$

$$= 7.55 \times 10^9 \text{ mm}^4.$$

Hence $I_{ef} = I_{cr} + (I_g - I_{cr}) \left(\frac{M_{cr}}{M_s} \right)^3$

$$I_{ef} = 7.55 \times 10^9 + (3.48 \times 10^{10} - 7.55 \times 10^9) \times \left(\frac{1113}{1250} \right)^3$$

$$I_{ef} = 2.66 \times 10^{10} \text{ mm}^4$$

short-term deflection

$$\Delta_{ST} = \frac{5}{384} \frac{((w_s^d - w_p) \cdot L^4)}{E_c I_{ef}}$$

$$w_p = \frac{8 \cdot p_e \cdot h}{L^2} = \frac{8 \times 1540 \times 0.396}{20^2} = 12.6 \text{ kN/m} \uparrow$$

$$\Delta_{ST} = \frac{5}{384} \times \frac{(25 - 12.6) \times 20,000^4}{33,800 \times 2.66 \times 10^{10}}$$

$$= 28.7 \text{ mm} \leq 29 \text{ mm} \quad \downarrow$$

(iii) additional long-term deflection.

$$\Delta_{LT} = \Delta_{pg} \cdot K_{cs} \quad \text{Cl 8.5.3.2}$$

$$K_{cs} = 2.0$$

$$\omega_{pg} = 14.3 - 12.6 = 1.7 \text{ mm} \downarrow$$

$$\Delta_{pg} = \frac{5}{384} \times \frac{1.7 \times 20,000^4}{33,800 \times 2.66 \times 10^{10}}$$

$$= 3.9 \text{ mm} \downarrow$$

$$\Delta_{LT} = 3.9 \times 2 = 7.8 \text{ mm} \downarrow$$

(iv) total long-term deflection.

$$\Delta_{\text{initial}} = (\Delta_g + \phi_s \Delta_a) - \Delta_p$$

$$= 29 \text{ mm} \downarrow$$

$$\Delta_{LT} = 7.8 \text{ mm}$$

$$\Delta_{\text{total}} = \Delta_{\text{initial}} + \Delta_{LT}$$

$$= 29 + 7.8 = 36.8 \text{ mm} \downarrow < \frac{L}{250} \therefore \text{ok}$$

Problem 7 - Solution - Serviceability design

⇒ Section Properties

$$A_g = 320,000 \text{ mm}^2 \quad I_g = 3.86 \times 10^{10} \text{ mm}^4$$

⇒ Prestress

$$P_j = 10 \times 184 \times 0.7 = 1288 \text{ kN}$$

Assume immediate losses = 5%

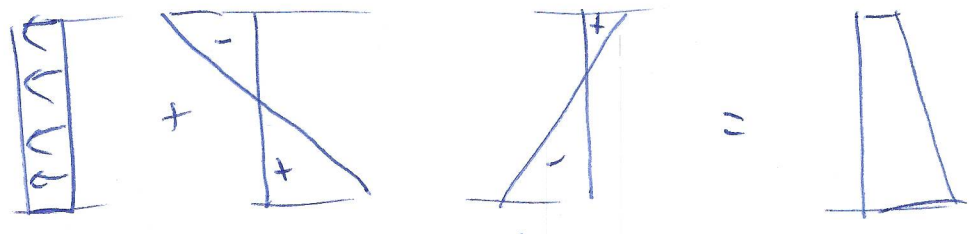
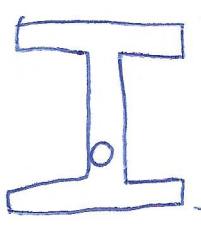
$$P_j \xrightarrow{5\% \text{ loss}} P_i = 0.95 \times 1288 = 1224 \text{ kN}$$

① Check adequacy of beam at transfer

@ transfer beam self-weight considered

$$W_g = 24 \text{ kN/m}^3 \times 0.32 \text{ m}^2 = 7.68 \text{ kN/m}$$

$$\text{Bending moment, } M_g = \frac{W_g L^2}{8} = \frac{7.68 \times 16^2}{8} = 246 \text{ kNm}$$


$$\frac{P}{A_g} + \frac{P_i \cdot e \cdot y_b}{I_g} - \frac{M_g \cdot y}{I_g} = \text{Resultant stress}$$

$$\sigma_b = \frac{P_i}{A_g} + \frac{P_i \cdot e \cdot y_b}{I_g} - \frac{M_g \cdot y_b}{I_g}$$
$$= \frac{1224 \times 10^3}{320,000} + \frac{1224 \times 10^3 \times 350 \times 800}{3.86 \times 10^{10}} - \frac{246 \times 10^6 \times 500}{3.86 \times 10^{10}}$$

$$= 3.9 + 5.9 - 3.3$$

$$= 6.5 \text{ MPa (compression)}$$

$$0.6 f_{cp} = 15 \text{ MPa} > \sigma_b (6.5) \therefore \text{OK}$$

② Total short-term deflection @ transfer

Equivalent upward load due to prestress

$$w_p = \frac{8 \cdot P_i h}{L^2} = \frac{8 \times 1224 \times 0.35}{16^2} = 13.4 \text{ kN/m} \uparrow$$

$$w_g = 7.68 \text{ kN/m} \downarrow$$

$$\text{Resultant load} = 7.68 \downarrow - 13.39 \uparrow = 5.71 \text{ kN/m} \uparrow$$

$$\Delta_{st} = \Delta_{py} = \frac{5 w_{net} L^4}{384 E_c I_g} = \frac{5 \times 5.71 \times 16,000^4}{384 \times 26,200 \times 3.86 \times 10^{10}} = 4.73 \text{ mm} \uparrow$$

$$\text{Take } E_c = 26,200 \text{ MPa}$$

$$T_b (3.1.2) d A_s 3600$$

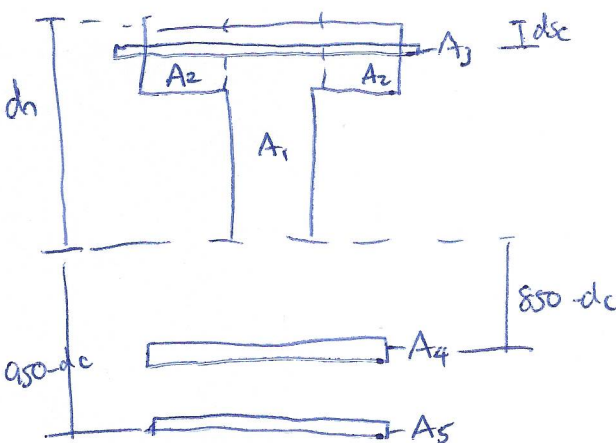
③ Total short-term deflection under service loads.

$w^* = 50 \text{ kN/m}$, Section is cracked as determined in Problem 5.

Need to perform cracked section analysis, calculate d_n and I_{cr}

$\Rightarrow$ Determine d_n by equating moments above & below neutral axis.

Consider transformed section, assume d_n in web



$$A_1 = 200 d_n \text{ mm}^2$$

$$A_2 = 400 \times 150 = 60,000 \text{ mm}^2$$

$$A_3 = A_{sc}(n-1) = 1800 \times \left[\frac{200,000}{28,600} - 1 \right] = 10787 \text{ mm}^2$$

$$A_4 = n A_p = \frac{190,000}{28,600} \times 1000 = 6643 \text{ mm}^2$$

$$A_5 = n A_{st} = \frac{200,000}{28,600} \times 3600 = 25175 \text{ mm}^2$$

$$\text{Take } E_c = 28,600 \text{ MPa} \cdot \cdot \cdot$$

$n = \text{modular ratio}$

$$\sum M @ NA (\sum A y = 0)$$

$$\left[200 d_n \times \frac{d_n}{2} \right] + \left[60,000 \times (d_n - 75) \right] + \left[10787 \times (d_n - 50) \right] - \left[6643 \times (850 - d_n) \right] - \left[25175 \times (950 - d_n) \right] = 0$$

$$100 d_n^2 + 102605 d_n - 34602150 = 0$$

$$\therefore d_n \approx 268 \text{ mm}$$

$\Rightarrow$ Determine second moment of area of cracked section about neutral axis $(\sum I + A \bar{y}^2)$

$$I_{ce} = \left[\frac{1}{12} \times 200 \times d_n^3 + 200 \times d_n \times \left(\frac{d_n}{2} \right)^2 \right] + \left[\frac{1}{12} \times 400 \times 150^3 + 400 \times 150 \times (d_n - 75)^2 \right] + \left[10787 \times (d_n - 50)^2 \right] + \left[6643 \times (850 - d_n)^2 \right] + \left[25175 \times (950 - d_n)^2 \right]$$

Substitute $d_n = 263 \text{ mm}$

$$I_{ce} = 1.85 \times 10^{10} \text{ mm}^4$$

$$I_{ef} = I_{ce} + (I_g - I_{ce}) \left(\frac{M_{cr}}{M_s} \right)^3 \quad \text{— Branson's formula.}$$

$$M_{ce} = \frac{629}{1.52} \text{ kNm}$$

$$\leftarrow [f_e = 1224 \times 0.8 = 979 \text{ kN}, \sigma_{cs} = 2.69]$$

$$M_s = \frac{50 \times 16^2}{8} = 1600 \text{ kNm}$$

$$\therefore I_{ef} = 1.85 \times 10^{10} + (3.86 \times 10^{10} - 1.85 \times 10^{10}) \left(\frac{629}{1600} \right)^3 = 1.97 \times 10^{10} \text{ mm}^4$$

$\Rightarrow$ Equivalent upward load due to prestress @ service load

$$w_p = \frac{8 f_e h}{L^2} = \frac{8 \times 978 \times 0.35}{16^2} = 10.7 \text{ kN/m} \uparrow$$

$$\Delta_{service} = \frac{5 w_{net} L^4}{384 E_c I_{ef}} = \frac{5 \times (50 - 10.7) \times 1600^4}{384 \times 28,600 \times 1.97 \times 10^{10}}$$

$$= 59.5 \text{ mm} \downarrow$$

(3)

④ Creep deflection

$$\Delta_{\text{creep}} = K_{cs} \Delta_{pg}$$

$$\text{where } K_{cs} = \left[2 - 1.2 \times \left(\frac{A_{sc}}{A_{st}} \right) \right] = 2 - 1.2 \left(\frac{1800}{3600} \right) = 1.4$$

$$\begin{aligned} \Delta_{\text{creep}} &= 1.4 \times (-473 \text{ mm } \uparrow) \\ &= \underline{6.6 \text{ mm } \uparrow} \end{aligned}$$

⑤ Total deflection

$$\Delta_{\text{total}} = \Delta_{\text{service}} + \Delta_{\text{creep}}$$

$$= 59.5 - 6.9 = 52.6 \text{ mm}$$

$$\text{allowable limit } \frac{L}{250} = \frac{16,000}{250} = 64 \text{ mm}$$

$$52.6 < 64 \therefore \text{OK.}$$